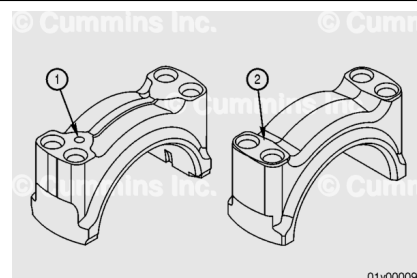


Trainings ▾

General Information

⚠ CAUTION ⚠

The fracture-split connecting rod requires a different large end rod bearing than the saw cut connecting rod. The bearings are specific to the connecting rod design and are not backward compatible. The oil drilling in the connecting rod and upper bearing shell will not be aligned if the wrong bearing is used. If the bearings are mismatched with the rod design, severe engine damage could occur.



⚠ CAUTION ⚠

To prevent damage to the fracture-split connecting rod, do not set the connecting rod or rod cap on the fracture-split connection. This can cause polishing and damage to the mating surface.

There are two styles of connecting rod assemblies on the ISX15 and QSX15 engines.

Newer model engines use a fracture-split (1) connection between the connecting rod and connecting rod cap.

Older model engines use a saw cut (2) connection between the connecting rod and connecting rod cap.

It is important to understand if the rod has a fracture-split or a saw cut connection before continuing with this procedure. The fracture-split connecting rod will have a visible hole drilled in the cap.

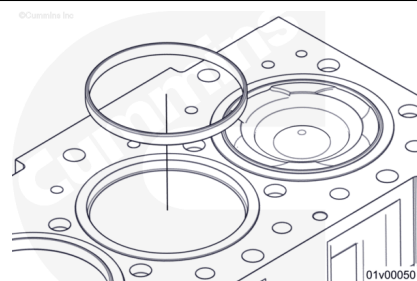
Note : Saw cut connecting rod assemblies may be mixed with fracture-split connecting rod assemblies within the same engine.

Connecting rods with a fracture-split surface **must** be treated with additional caution. The two pieces of the connecting rod can **not** be rubbed together. This will damage the mating surfaces. Use care to **not** drop either piece of the connecting

rod. Fracture-split connecting rods **must only** be handled if the two pieces of the connecting rod are tightened to the correct specification, or completely separated.

⚠ CAUTION ⚠

Antipolishing ring cylinder liners are available for production and service on 15L engines. These liners are not compatible with all piston designs. If an antipolishing ring cylinder liner is used in conjunction with a high top ring piston, there will be a mechanical interference. Interference will prevent assembly of the engine.



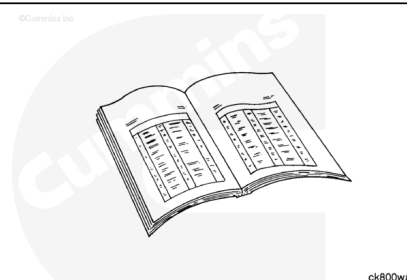
15L engines started using antipolishing ring cylinder liners for production and field repairs December 2014.

It is important to identify which cylinder liner design the engine contains in order to repair the engine correctly with compatible components. See Service Bulletin 4383774 for more information on antipolishing ring liners.

Preparatory Steps

⚠ WARNING ⚠

To reduce the possibility of personal injury, reduce the possibility of direct contact of hot oil with your skin.



⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Reduce the possibility of inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam

can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

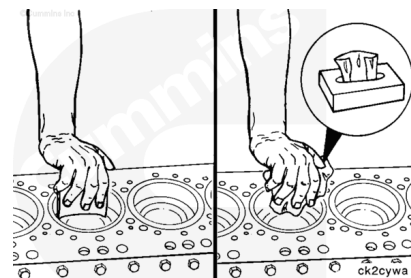
This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this component.

- Identify the power cylinder component. See Service Bulletin 4383774.
- Drain the lubricating oil. Refer to Procedure 007-037 in Section 7 (</qs3/pubsys2/xml/en/procedures/10/10-007-037-tr.html>).
- Drain the cooling system. Refer to Procedure 008-018 in Section 8 (</qs3/pubsys2/xml/en/procedures/10/10-008-018-tr.html>).
- Remove the lubricating oil pan. Refer to Procedure 007-025 in Section 7 (</qs3/pubsys2/xml/en/procedures/10/10-007-025-tr.html>).
- Remove the lubricating oil suction tube. Refer to Procedure 007-035 in Section 7 (</qs3/pubsys2/xml/en/procedures/10/10-007-035-tr.html>).
- Remove the block stiffener plate. Refer to Procedure 001-089 in Section 1 (</qs3/pubsys2/xml/en/procedures/10/10-001-089-tr.html>).
- Remove the piston cooling nozzles. Refer to Procedure 001-046 in Section 1 (</qs3/pubsys2/xml/en/procedures/10/10-001-046-tr.html>).
- Remove the connecting rod cap. Refer to Procedure 001-005 in Section 1 (</qs3/pubsys2/xml/en/procedures/10/10-001-005-tr.html>).
- Remove the cylinder head. Refer to Procedure 002-004 in Section 2 (</qs3/pubsys2/xml/en/procedures/10/10-002-004-tr.html>).

Remove

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

⚠ CAUTION ⚠

Do not use emery cloth or sandpaper to remove carbon from the cylinder liners. Aluminum oxide or silicon particles from emery cloth or sandpaper can cause serious engine damage. Do not use any abrasives in the ring travel area. The cylinder liner can be damaged.

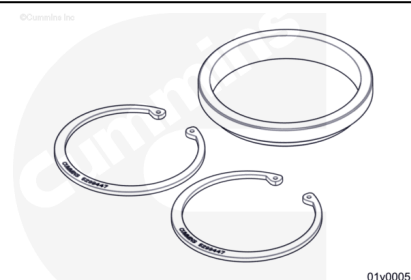
Use a fine fibrous abrasive pad such as Scotch-Brite™ 7448, Part Number 3823258, or equivalent, and solvent to remove the carbon.

For antipolishing ring cylinder liners, it is **not** necessary to remove carbon prior to removing the cylinder kit. Remove the antipolishing ring.

For antipolishing ring cylinder liners, the antipolishing ring **must** be removed prior to removing the piston and connecting rod assemblies. In most cases, the antipolishing ring can be removed by hand.

Use snap rings, Part Number 5299447, from service tool kit, Part Number 5299448, if antipolishing rings can **not** be removed by hand.

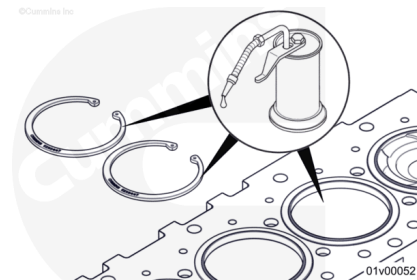
To remove tight antipolishing rings, rotate the crankshaft until the piston crown of interest is approximately 100 mm [4 in] below the antipolishing ring. Use the engine barring device.



Note : If reusing liners, apply a coating of clean engine oil to the snap rings and liner bore. This prevents damage to the bore during installation of the snap ring.

Install one snap ring in the cylinder to be removed. Position the snap ring in the bore between the piston crown and antipolishing ring.

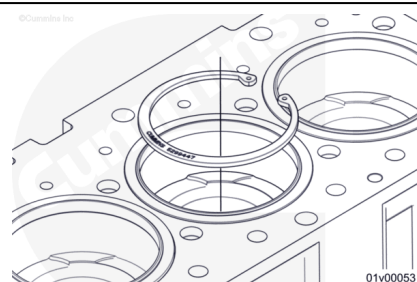
Note : For the removal of multiple cylinders, install one snap ring in each of the two cylinders in the same crank position (1 and 6, 2 and 5, or 3 and 4).



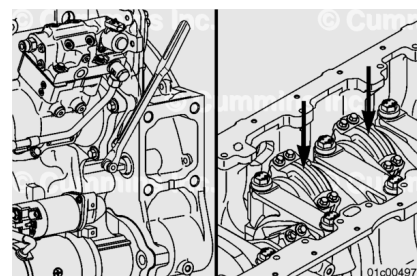
Bar the engine over. Bring the piston to top dead center. When the piston comes in contact with the snap ring, it will align the snap ring and drive it into the bottom face of the antipolishing ring. Continuing to bar the engine over will press the antipolishing ring out of the liner counterbore.

Repeat for all additional cylinders requiring removal.

If reusing the antipolishing rings, label each ring with the cylinder location from which it was removed.

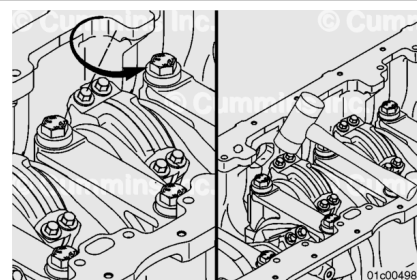


Use the engine barring device to rotate the crankshaft so that two of the connecting rods are at bottom dead center.



⚠ CAUTION ⚠

Use care not to damage the fracture-split mating face(s) on the connecting rod and connecting rod cap while the connecting rod caps are removed. If the fracture-split surface is damaged, the connecting rod and connecting rod cap must be replaced to reduce the possibility of engine damage. Incorrect assembly can also damage the connecting rod and connecting rod cap.



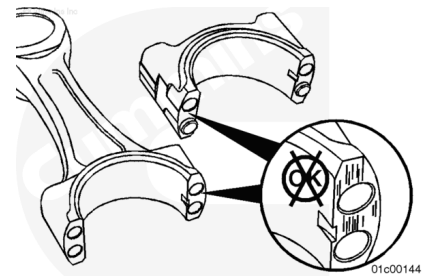
Do **not** remove the capscrews at this time.

Loosen the connecting rod capscrews.

Hit the connecting rod capscrews with a rubber hammer to loosen the connecting rod caps from the dowels.

⚠ CAUTION ⚠

To prevent damage to the fracture-split connecting rod and connecting rod cap, do not set the connecting rod or connecting rod cap on the fracture-split mating face(s). This can cause polishing and damage to the mating surface(s).



Note : Minimize the number of joint interactions between the connecting rod and rod cap. If the fracture-split mating surface becomes damaged or worn, the rod cap will **not** fit as designed to the connecting rod and engine damage can occur.

Remove the capscrews and the rod cap. Note the corresponding rod and rod cap as well as the orientation of the rod cap.

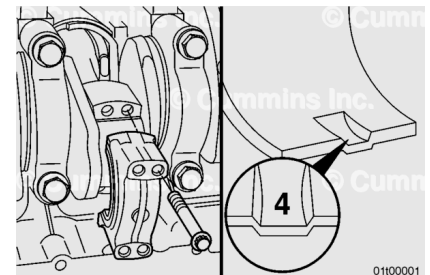
Use care **not** to damage the fracture-split connection.

Do **not** set the rod cap down on the fracture-split face.

Note : Both the connecting rod cap and the connecting rod are serialized on one face to make sure of proper orientation.

Connecting rod shell bearing upper or lower location is denoted by UPPER or UPR (for connecting rod) and LOWER or LWR (for connecting rod cap) stamped into the back of the bearing shells. The upper connecting rod shell bearing can also be identified by the oil port in the bearing.

Remove the bearing shell from the rod cap and mark the cylinder number and the letter "L" on the flat surface of the bearing tangs.

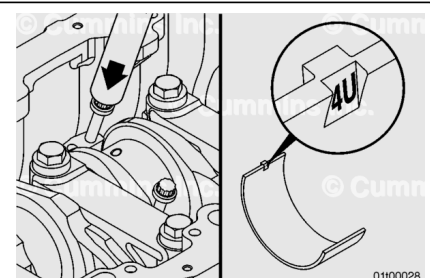


Install two of the capscrews back into the rod finger-tight or install connecting rod guide pins, Part Number 5298634.

Use a T-handle piston pusher. Push **only** against the head of the capscrew, to push the rod up far enough to allow the upper bearing shell to be removed.

Do **not** push against the fracture-split face of the rod.

Use caution **not** to scratch or damage the journal with the rod, capscrew, or piston pusher.

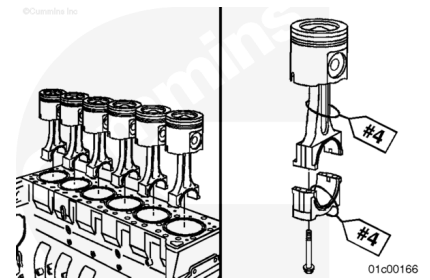


Remove the bearing shell and mark the cylinder number and the letter U (upper) on the flat surface of the bearing tang.

Use both hands to remove the piston and connecting rod assembly.

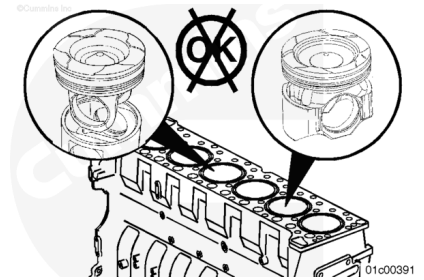
The piston and connecting rod assemblies **must** be installed in the same cylinder from which they were removed to provide for proper fit of worn mating surfaces if parts are reused.

Use a tag to mark the cylinder number from which each piston and connecting rod assembly was removed.



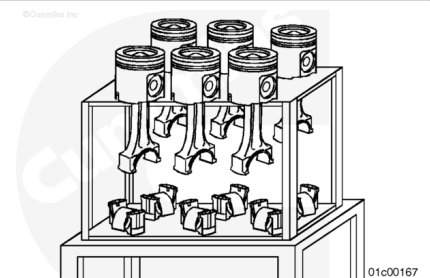
⚠ CAUTION ⚠

To reduce the possibility of engine damage, all pistons within a given engine must be of the same design - all are to be either the articulated design or the single-piece design. Do not mix piston designs in the same engine. Single-piece pistons must have drilled rods and connecting rod bearings. Articulated pistons can use either drilled or non-drilled rods and bearings.



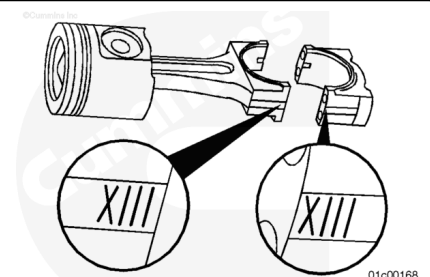
⚠ CAUTION ⚠

Connecting rods with a fracture-split surface must be treated with caution. The two pieces of the connecting rod cannot be rubbed together. This will damage the mating surfaces. Use care to not drop either piece of the connecting rod. Fracture-split connecting rods must only be handled if the two pieces of the connecting rod are tightened to the correct specification, or are completely separated.



Place the connecting rod and piston assemblies in a container to protect them from damage.

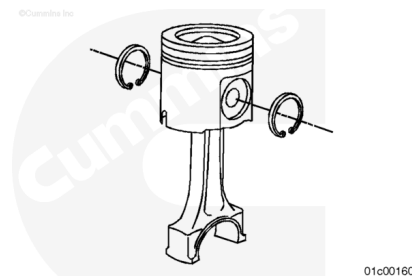
A unique number (**not** cylinder number) is stamped on the connecting rod and matching connecting rod cap. When the connecting rods and connecting rod cap are installed in the engine, the numbers on the connecting rod and cap **must** match and be installed on the same side (the exhaust side) of the engine.



Disassemble

Articulated Piston

Use internal snap ring pliers to remove the snap rings from both sides of the piston.

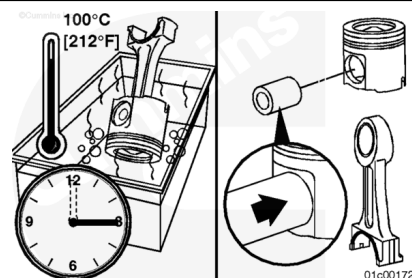


⚠ CAUTION ⚠

Do not use a hammer to remove the piston pins. The piston can distort, causing it to seize in the liner.

If the piston pin can **not** be easily removed by hand, place the piston and connecting rod assembly in a container of water. Heat the piston in boiling water for 15 minutes.

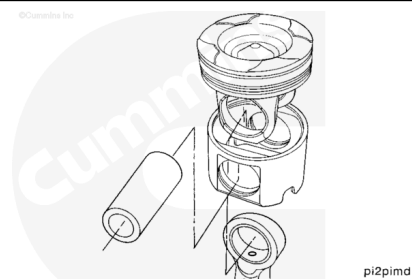
Use a blunt tool to push the piston pin from the piston and connecting rod assembly.



⚠ CAUTION ⚠

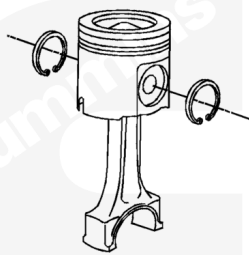
When the piston pin is removed from an articulated piston, the skirt will separate from the crown. Use care to prevent damage to the piston.

If they are to be reused, mark the number of the cylinder from which the piston, crown, skirt, and pin were removed on the parts to make sure they are installed in the correct cylinder.



Single-Piece Piston

Use internal snap ring pliers to remove the snap rings from both sides of the piston.



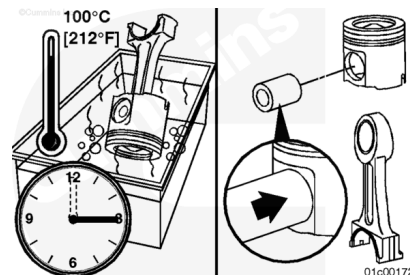
01c00160

⚠ CAUTION ⚠

Do not use a hammer to remove the piston pins. The piston can distort, causing it to seize in the liner.

If the piston pin can **not** be easily removed by hand, place the piston and connecting rod assembly in a container of water. Heat the piston in boiling water for 15 minutes.

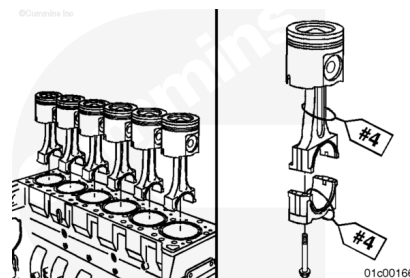
Use a blunt tool to push the piston pin from the piston and connecting rod assembly.



01c00172

The piston and connecting rod assemblies **must** be installed in the same cylinder from which they were removed to provide for proper fit of worn mating surfaces if parts are reused.

Use a tag to mark the cylinder number from which each piston and connecting rod assembly was removed.



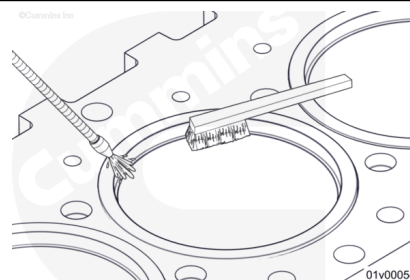
01c00166

Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

Inspect the antipolishing ring liner counterbore for carbon or debris that can interfere with seating of the antipolishing ring during final assembly.



01v00054

Clean the counterbore. Use a cleaning solvent, Part Number 5298527, or equivalent. Use a brass wire brush to clean carbon deposits.

Refer to Procedure 001-028 in Section 1 (</qs3/pubsys2/xml/en/procedures/10/10-001-028-tr.html>) for the reuse inspection of the antipolishing ring and cylinder liner.

Assemble

Articulated Piston

All pistons (bushed and nonbushed) **must** be properly lubricated prior to assembly.

For all pistons, use a clean, lint-free cloth to apply a heavy film of 85W-140 gear oil with EP additive to the following locations:

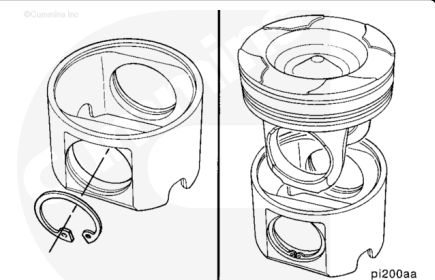
- Piston pin bore inside diameter in the piston crown
- Piston pin bore inside diameter in the piston skirt
- Rod bushing inside diameter
- Piston pin.



Assemble the piston crown into the piston skirt.

⚠ CAUTION ⚠

The retainer snap ring must be seated completely in the piston pin groove to prevent engine damage during engine operation.



It is **not** necessary to heat the articulated pistons before assembly. The piston pin is slip-fitted.

The crown and skirt can **only** be assembled in one position.

Install a new snap ring into one piston pin bore of each piston skirt.

If the pistons are being reused, the crown, skirt, and pin **must** be matched as they were when they were removed.

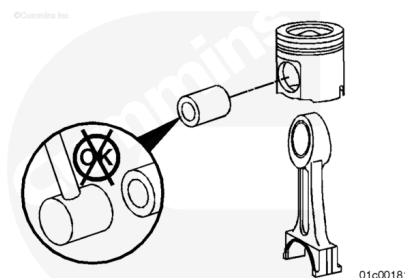
Position the skirt over the piston crown.

⚠ CAUTION ⚠

Do not use a hammer to install the piston pin. The piston can distort, causing it to seize in the liner.

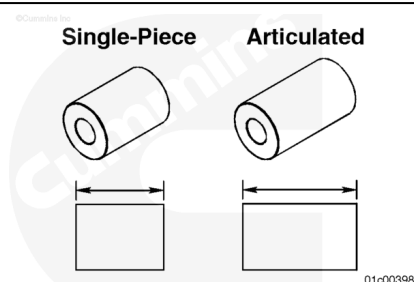
⚠ CAUTION ⚠

The lock tang on the connecting rod must be in line with the cooling nozzle notch in the piston skirt and on the side opposite the deep valve pocket of the piston crown. Failure to follow this step will cause extensive engine damage.



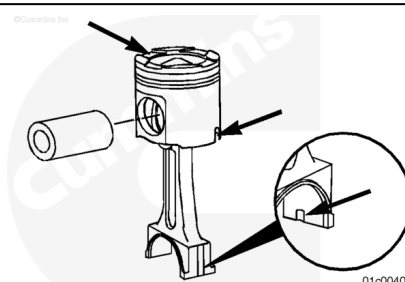
⚠ CAUTION ⚠

The piston pin for the articulated piston is longer in length than the pin for single-piece pistons. Do not use articulated piston pins in single-piece pistons. Likewise, do not use single-piece pins in articulated pistons.



Align the pin bore of the connecting rod with the pin bore of the piston skirt and crown, and install the piston pin.

On articulated pistons, make sure the locking tang on the connecting rod is on the same side as the cooling nozzle notch in the piston skirt and opposite the deep valve pocket on the crown.

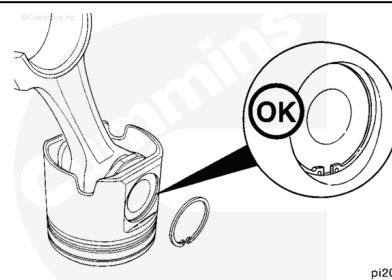


⚠ CAUTION ⚠

The snap ring must be seated completely in the piston groove to prevent engine damage during engine operation.

Install a new snap ring into the piston pin bore.

Install piston rings. Refer to Procedure 001-047 in Section 1 (</qs3/pubsys2/xml/en/procedures/10/10-001-047-tr.html>).



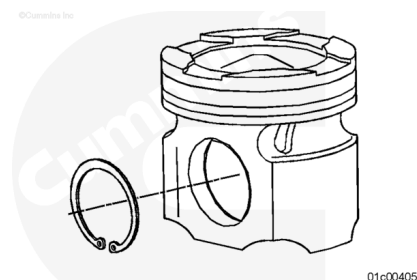
Single-Piece Piston

⚠ CAUTION ⚠

The retainer snap ring must be seated completely in the piston pin groove to prevent engine damage during engine operation.

Install a new snap ring into one piston pin bore of each single-piece piston.

If the pistons are being reused, the single-piece piston and pin **must** be matched as they were when they were removed.

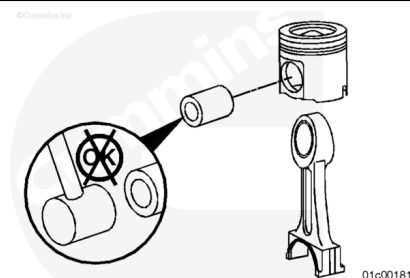


⚠ CAUTION ⚠

Do not use a hammer to install the piston pin. The piston can distort, causing it to seize in the liner.

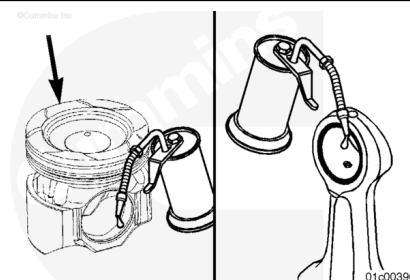
⚠ CAUTION ⚠

The lock tang on the connecting rod must be in line with the cooling nozzle notch in the piston skirt and on the side opposite the deep valve pocket of the piston crown. Failure to follow this step will cause extensive engine damage.



⚠ CAUTION ⚠

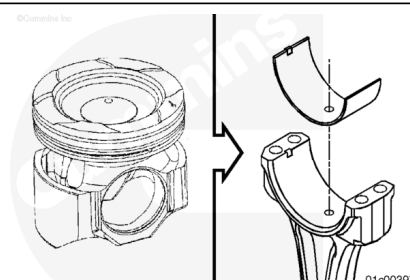
Generously coat both single-piece piston pin bores with a heavy weight gear oil, such as 85W-140 EP, when assembling single-piece pistons. Also lubricate the connecting rod bushing. If gear oil is not available, Lubriplate™, Part Number 3163086, is permissible. Failure to adequately lubricate the pin joint can result in extensive engine damage, particularly with single-piece pistons.



Use clean 85W-140 oil to coat the piston pin.

⚠ CAUTION ⚠

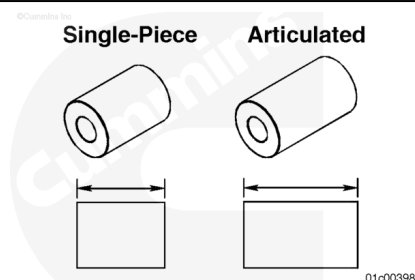
The fracture-split connecting rod requires a different large end rod bearing from the saw cut connecting rod. An engine may have a mixture of fracture-split and saw cut connecting rods, so it is critical to install the correct rod bearings in the correct connecting rod design. Failure to do so could result in severe engine damage.



The single-piece piston requires a drilled connecting rod and a drilled upper connecting rod bearing to lubricate the pin joint. Do **not** assemble a single-piece piston with a non-drilled connecting rod nor a non-drilled connecting rod bearing. Make certain when assembling a single-piece piston that the lubricating hole in the connecting rod bearing lines up with lubricating hole in the connecting rod.

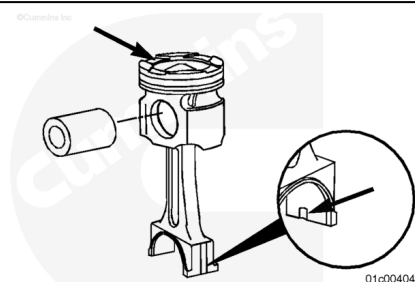
⚠ CAUTION ⚠

The piston pin for single-piece pistons is shorter in length than the pin for articulated pistons. Do not use single-piece piston pins in articulated pistons. Likewise, do not use articulated pins in single-piece pistons.



On single-piece pistons, the locking tang on the connecting rod is to be on the opposite side of the deep valve pocket.

Install the piston pin.



⚠ CAUTION ⚠

The snap ring must be seated completely in the piston groove to prevent engine damage during engine operation.

Install a new snap ring into the piston pin bore.

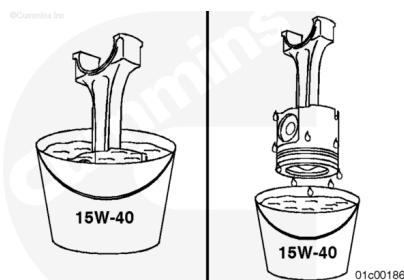
Install the piston rings. Refer to Procedure 001-047 in Section 1. (</qs3/pubsys2/xml/en/procedures/10/10-001-047-tr.html>)



Install

Immerse the piston in a container of clean 15W-40 oil.

Remove the piston from the container. Let the excess oil drain from the piston.



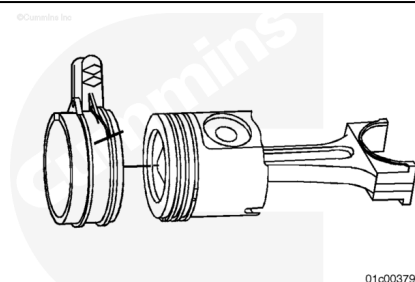
Use piston ring compressor, Part Number 3164846, to compress the rings. Refer to Procedure 001-047 in Section 1. (</qs3/pubsys2/xml/en/procedures/10/10-001-047-tr.html>)

When reusing piston rings, be sure the rings are matched up with the same piston from which they were removed.

Inspect the cylinder liners for reuse. Refer to Procedure 001-028 in Section 1. (</qs3/pubsys2/xml/en/procedures/10/10-001-028-tr.html>)

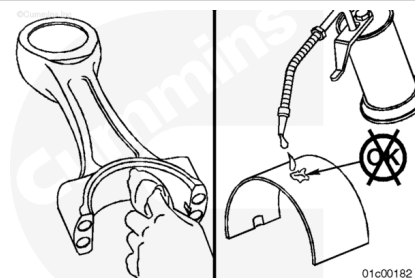
Use the following procedure when installing new piston rings with used liners. Refer to Procedure 001-047 in Section 1. (</qs3/pubsys2/xml/en/procedures/10/10-001-047-tr.html>)

The cylinder block and all parts **must** be clean before assembly.



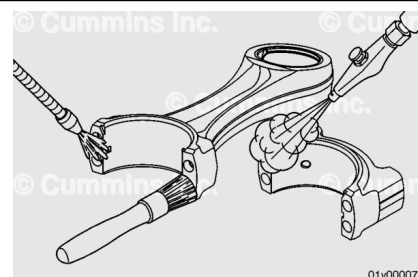
Use a clean, lint-free cloth to clean the connecting rods and bearing shells.

Do **not** lubricate the back side of the bearing shells. The operating clearance of the bearing will be reduced and the bearing can be damaged during engine operation.



Fracture-split Connecting Rods

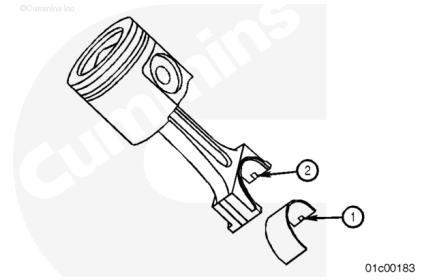
- The fracture joint may produce debris at each separation, so it is important to clean before reassembly. Use solvent and a nylon bristle brush to clean the contact surface between the connecting rod and the connecting rod cap. Dry the contact surface with compressed air.



If new bearings are **not** used, the used bearings **must** be installed on the same connecting rod from which they were removed.

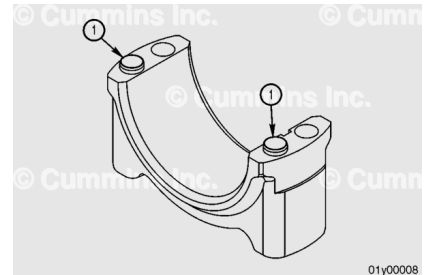
Install the upper bearing shell into the connecting rod.

The tang of the bearing shell (2) **must** be in the slot of the cap (1). The end of the bearing shell **must** be even with the cap mounting surface.



Saw Cut Connecting Rods

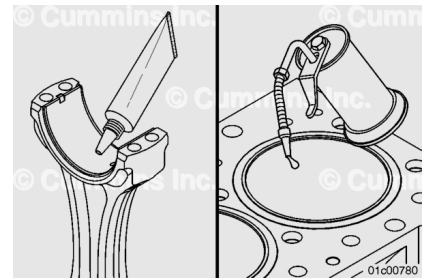
- Verify both connecting rod dowel rings are installed in either the connecting rod or connecting rod cap.



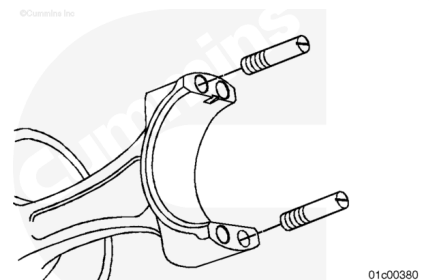
Use Lubriplate™ 105, Part Number 3163086, or equivalent, to coat the entire inside circumference of the bearing shell.

Apply a film of clean 15W-40 oil to the cylinder liner.

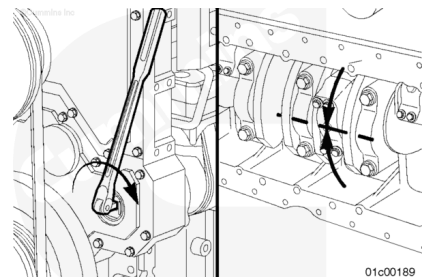
For antipolishing ring cylinder liners, apply a film of clean 15W-40 oil to the cylinder liner counterbore.



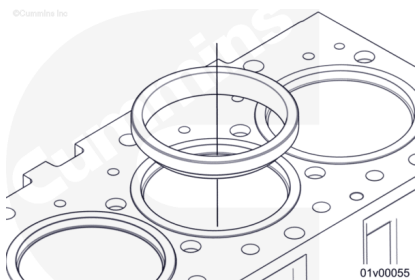
Install connecting rod guide pins, Part Number 5298634.



Use the compressor drive or barring device to rotate the crankshaft so the connecting rod journal of the connecting rod being installed is at bottom dead center.



Install the piston ring compressor antipolishing ring adapter, Part Number 5299339, from service tool kit, Part Number 5299448, into the cylinder liner antipolishing ring counterbore.



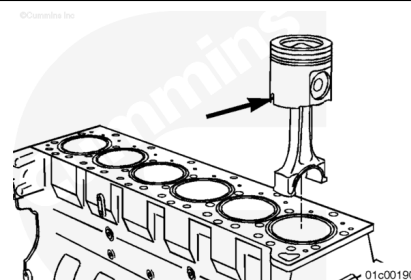
⚠ CAUTION ⚠

Failure to align the deep valve pocket and the piston skirt cooling nozzle notch correctly will result in extensive engine damage.

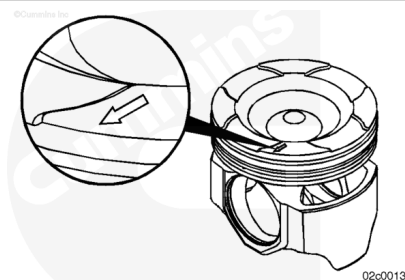
Align the piston crown deep valve pocket on the exhaust side of the engine, and the piston skirt cooling nozzle notch to the intake side of the engine.

If the deep valve pocket and the piston skirt cooling nozzle notch are **not** on opposite sides, the piston and connecting rod assembly is **not** assembled correctly. Disassemble the piston and connecting rod and assemble correctly.

Insert the connecting rod through the cylinder liner until the ring compressor contacts the top of the liner.



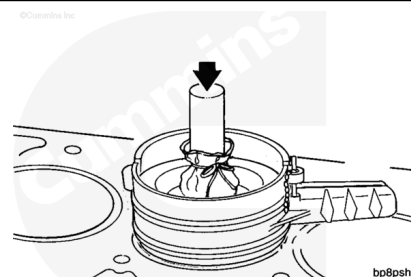
Both articulated pistons and single-piece pistons have an arrow on the top of the piston crown which will point to the front of the engine when the piston is installed properly.



⚠ CAUTION ⚠

Do not use a metal drift to push the piston into the cylinder liner. The piston rings or cylinder liner can be damaged.

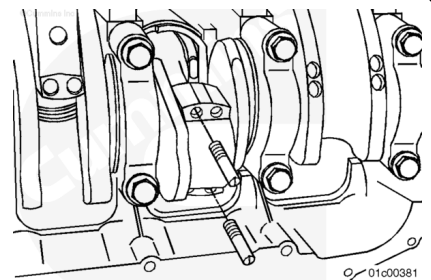
Hold the ring compressor against the cylinder liner or antipolishing ring adapter. Push the piston through the ring compressor and into the cylinder liner. Push the piston until the top ring is completely in the cylinder liner.



If the piston does **not** move freely, remove the piston and inspect for broken or damaged rings.

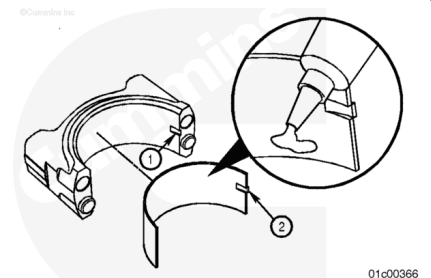
Use the nylon guide pins to align the connecting rod with the crankshaft while pushing the piston and connecting rod assembly into place.

Remove the nylon guide pins.



⚠ CAUTION ⚠

The fracture-split connecting rod requires a different large end rod bearing from the saw cut connecting rod. An engine may have a mixture of fracture-split and saw cut connecting rods, so it is critical to install the correct rod bearings in the correct connecting rod design. Failure to do so could result in severe engine damage.

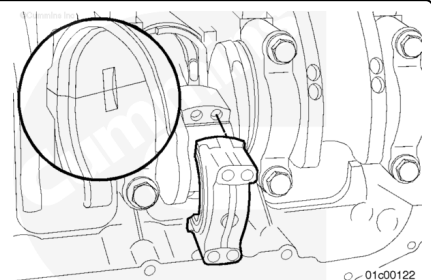


Install the bearing shell into the connecting rod cap with the tang (2) of the bearing in the slot (1) of the cap.

Use clean Lubriplate™ 105 multi-purpose lubricant, or equivalent, to lubricate the bearing shell to the crankshaft journal mating surface and the connecting rod capscrews.

⚠ CAUTION ⚠

The rod cap alpha characters must match the alpha characters on the connecting rod and must be installed with the characters aligned to prevent damage to the connecting rods and crankshaft. The locking tang of the connecting rod cap must face toward the intake side of the cylinder block. Rod and cap mating surfaces must be clean or engine damage can occur.



Install the cap onto the connecting rod.

Lubricate the threads of the capscrews with clean engine oil.

Thread the capscrews into the connecting rod and hand tighten.

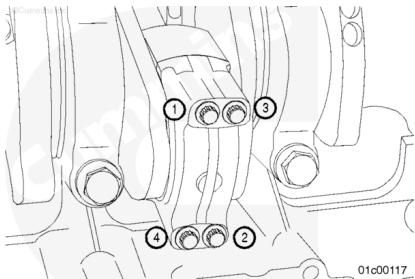
The procedure for tightening used capscrews is different than the procedure for tightening new capscrews.

Used Connecting Rod Capscrews

- 1. Hand-tighten the capscrews in numerical sequence.
- 2. Tighten the capscrews in numerical sequence: 70 N•m [52 ft-lb].
- 3. Check the capscrew torque specifcations in numerical sequence: 70 N•m [52 ft-lb].
- 4. Rotate the capscrews in numerical sequence 60 degrees.

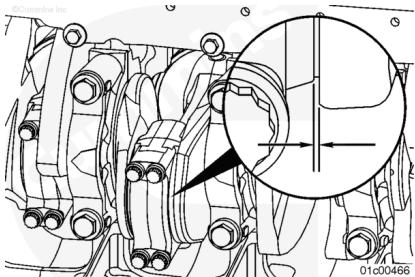
New Connecting Rod Capscrews

- 1. Hand-tighten the capscrews in numerical sequence.
- 2. Tighten the capscrews in numerical sequence: 40 N•m [30 ft-lb].
- 3. Loosen the capscrews in numerical sequence.
- 4. Tighten the capscrews in numerical sequence: 70 N•m [52 ft-lb]
- 5. Check the capscrew torque specification in numerical sequence: 70 N•m [52 ft-lb].
- 6. Rotate the capscrews in numerical sequence 60 degrees.



Measure the connecting rod side clearance.

Connecting Rod Side Clearance		
mm		in
0.18	MIN	0.007
0.48	MAX	0.019



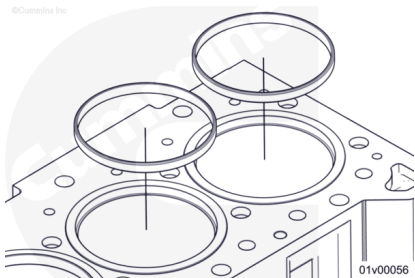
Note : The connecting rod **must** move freely from side-to-side on the crankshaft journal. If the connecting rod does **not** move freely, remove the connecting rod cap and make sure the bearing shells are the correct size. Check for dirt or damage on the crankshaft and the bearing shells.

Repeat the above steps to install the remaining bearing shells and connecting rod caps.

Verify all antipolishing ring liner counterbores are clean and free of debris.

Install the antipolishing rings into cylinder liner counterbores.

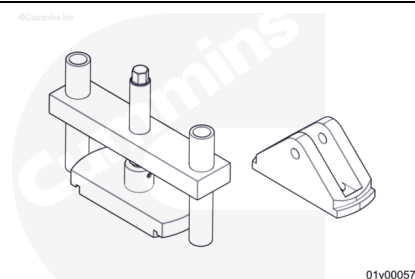
Note : The antipolishing ring is **not** directional. It can be installed in either orientation.



The majority of antipolishing rings will assemble by hand with no additional tooling.

For hard to install antipolishing rings, use the following service tools:

- Cylinder liner installation tool, Part Number 3164606
- Cylinder liner puller plate, Part Number 3162462.



Position the antipolishing ring on the top of the cylinder liner counterbore. Align rings so it sits squarely on top of the counterbore.

Install the cylinder liner puller plate, Part Number 3162462, in the center of the antipolishing ring. Install the cylinder liner puller plate with the bottom facing up for proper seating and alignment.

Install the cylinder liner installation tool, Part Number 3164606, over the antipolishing ring and puller plate.

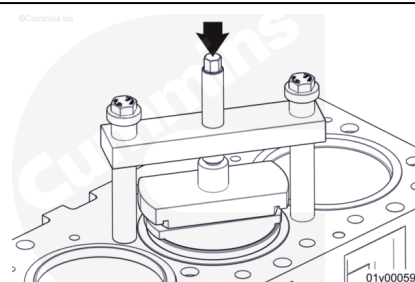
Thread the installation tool down slowly. Pressure the antipolishing ring into the liner counterbore.

If the antipolishing ring severely misaligns while pressing it into place, stop the installation process and inspect for damage.

If both the liner and antipolishing ring are undamaged, repeat the above steps.

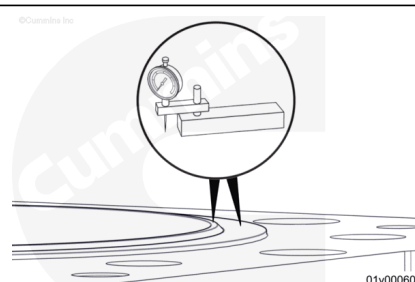
Refer to Procedure 001-028 in Section 1 (</qs3/pubsys2/xml/en/procedures/10/10-001-028-tr.html>) for reuse guidelines, including component dimensional checks, if after two attempts the antipolishing ring will not press.

Discard components that do **not** fall within the reuse specifications.



Measure antipolishing ring height to verify they are fully seated.

Measure the protrusion of the antipolishing ring from the top of the cylinder liner. Use depth gauge, Part Number 3164438.



If the ring is visually seated squarely in the counterbore, only one measurement of this height is necessary for each cylinder liner.

The antipolishing ring should **not** exceed the maximum height listed.

Maximum Antipolishing Ring Protrusion From the Cylinder Liner		
mm		in
1.7	MAX	0.067

If the measurement exceeds the maximum value, the antipolishing ring will interfere with the head gasket clamp load.

Remove the antipolishing ring. Inspect for debris in the liner counterbore or on the antipolishing ring.

Finishing Steps

⚠ WARNING ⚠

This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this component.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Reduce the possibility of inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.



ck800wa

Note : When rebuilding an engine with single-piece pistons that was originally built with articulated pistons, stamp the dataplate with the letters SP. This will alert future repair locations that this engine has been rebuilt with single-piece pistons.

Note : All engines must be run in after a rebuild or a repair involving the replacement of one or more piston ring sets, cylinder liners, or pistons. Reference Section 14, Engine Testing, for appropriate run-in procedure.

Note : Use the following procedure for general run in test overview. Refer to Procedure 014-999 in Section F. (</qs3/pubsys2/xml/en/procedures/10/10-014-999.html>)

- Install the connecting rod cap. Refer to Procedure 001-005 in Section 1 (</qs3/pubsys2/xml/en/procedures/10/10-001-005-tr.html>).
- Install the piston cooling nozzles. Refer to Procedure 001-046 in Section 1 (</qs3/pubsys2/xml/en/procedures/10/10-001-046-tr.html>).
- Install the block stiffener plate. Refer to Procedure 001-089 in Section 1 (</qs3/pubsys2/xml/en/procedures/10/10-001-089-tr.html>).
- Install the lubricating oil suction tube. Refer to Procedure 007-035 in Section 7 (</qs3/pubsys2/xml/en/procedures/10/10-007-035-tr.html>).
- Install the lubricating oil pan. Refer to Procedure 007-025 in Section 7 (</qs3/pubsys2/xml/en/procedures/10/10-007-025-tr.html>).
- Install the cylinder head. Refer to Procedure 002-004 in Section 2 (</qs3/pubsys2/xml/en/procedures/10/10-002-004-tr.html>).
- Prime the lubricating oil system. Refer to Procedure 007-037 in Section 7 (</qs3/pubsys2/xml/en/procedures/10/10-007-037-tr.html>).
- Fill the cooling system with coolant. Refer to Procedure 008-018 in Section 8 (</qs3/pubsys2/xml/en/procedures/10/10-008-018-tr.html>).
- Fill the engine with clean lubricating oil. Refer to Procedure 007-037 in Section 7 (</qs3/pubsys2/xml/en/procedures/10/10-007-037-tr.html>).

- Start and operate the engine to normal operating temperature and check for leaks.

Last Modified: 24-Apr-2015
